Chapter 15

Esophagus

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Epidemiology and clinical features

Key facts

- The commonest malignant esophageal tumors (Box 15.1) worldwide are *squamous cell carcinoma* and *adenocarcinoma*: adenocarcinoma is commonest in the USA where the incidence of squamous cell is falling.
- Most patients present with dysphagia to solids, and eventually liquids.
- 50% of patients have metastatic disease at the time of diagnosis.

Epidemiology and risk factors

- The commonest esophageal cancer in the West is adenocarcinoma. In the USA the incidence of adenocarcinoma is rising faster than the incidence of melanoma, prostate, breast, lung, and colorectal cancer.
- The incidence of squamous cell carcinoma is decreasing.
- In the USA, in 2010 there were an estimated 16 640 new cases of esophageal cancer, with 14 500 deaths (National Cancer Institute, 2010).

Squamous cell carcinoma

Most commonly occurs in the upper 2/3 of the esophagus.

- Smoking and alcohol consumption: synergistic (↑risk 44×).
- Nitrosamines: associated with pickled vegetables and cured meats.
- Diets deficient in vitamins A, C, and riboflavin.
- Previous SCCs of the head and neck.
- Previous external beam therapy.
- Chronic stasis injury:
  - Achalasia.
  - Esophageal webs.
  - Esophageal diverticulum.
- Chronic irritation: caustic burn injury, chronic ingestion of hot foods.
- Celiac disease.
- Tylosis.

Box 15.1 Etiology of esophageal tumors

- Common primary tumors (accounting for >95% of USA cases):
  - Adenocarcinoma.
  - Squamous cell carcinoma.
- Unusual primary tumors:
  - Adenoid cystic carcinoma.
  - Mucoepidermoid carcinoma.
  - Verrucous carcinoma.
  - Small-cell carcinoma.
  - Melanoma.
  - Lymphoma (1–2% of USA cases).
  - Mesenchymal tumors.
- Secondary tumors:
  - Breast.
  - Lung.
  - Stomach.

Adenocarcinoma

Most commonly found in the lower 2/3 of the esophagus.

- Barrett's metaplasia (see 📖 p780):
  - Due to chronic acid exposure (gastroesophageal reflux disease (GERD), 📖 p778).
  - Transformation of usual non-keratinizing stratified squamous epithelium to intestinal columnar epithelium with goblet cells.
  - Estimated annual incidence of malignant transformation 1–2%.
- Male gender:
  - 7× more likely to develop adenocarcinoma than SCC.
  - 3:1 male to female ratio.

Clinical features

- Most patients present with *dysphagia*, initially to solids, then progressing to liquids.
- With cervical lesions, patients often can point directly at point of obstruction, whereas with distal 2/3 lesions, patients have only a vague sense of the point of obstruction.
  - Odynophagia (pain on swallowing).
  - Weight loss.
  - Bleeding, anemia.
  - Aspiration, regurgitation, cough.

Diagnosis

- This is covered in more detail on 📖 pp760–763.
- Patients with dysphagia should undergo a barium esophagram to identify a stricture, and assess its site, length, and severity.
- Abnormalities on esophagram are evaluated with esophagogastroduodenoscopy (EGD). Brushings for cytology, and biopsy of suspicious lesions should be performed.
- Contrast CT of chest and upper abdomen allows accurate assessment of spread, and mediastinal lymph nodes but is less sensitive at assessing invasion into adjacent structures: PET is a useful adjunct to assess spread.

Imaging for diagnosis and staging

Key facts

- Accurate staging of esophageal cancer includes: (1) endoscopy, (2) chest and abdominal CT, (3) PET, and (4) EUS.
- EUS and CT together offer a 79% accuracy for T staging and 82% accuracy for N staging.

Chest and abdominal CT scan

Chest and abdominal contrast CT helps define locoregional spread, invasion of local structures, distant metastasis, and key anatomic structures.

Endoscopic ultrasound

- This is the gold standard for defining T stage: it allows detailed assessment of esophageal wall and surrounding structures.
- Also allows fine needle aspiration (FNA) of suspicious lymph nodes.
- EUS identifies five sonographic layers (Fig. 15.1):
  - Layers 1–2: mucosa.
  - Layer 3: submucosa.
  - Layer 4: muscularis propia.
  - Layer 5: adventitia.
- EUS has several limitations: it is poor at assessing tracheobronchial invasion, may be unable to pass through very tight stenoses, and cannot detect distant metastases.

PET scan

PET is most useful as a combination PET/CT fusion where it has a 78% sensitivity for identifying nodal disease and distant metastasis.

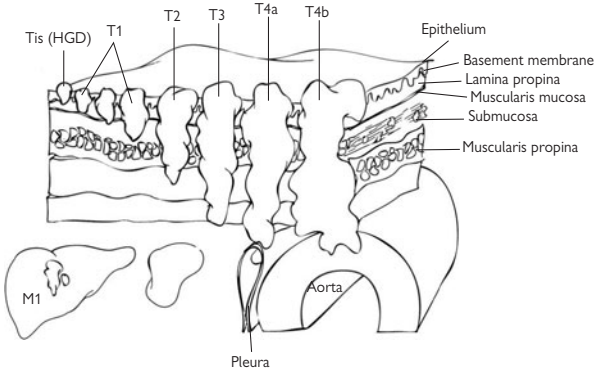


Fig.15.1 Layers of the esophageal wall, as they correspond to the five layers seen in an EUS image.

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Staging esophageal cancer

Key facts

See Tables 15.1–15.3 for stage grouping of different cancers.

- Stage 0, I, and IIA cancers are treated with esophagectomy, with postoperative chemoradiotherapy for postsurgical stage IIB and III cancers.
- Localized stage IIB and III are treated with trimodality therapy (📖 p772).

Table 15.1 Stage grouping for esophageal cancer

Primary tumor (T)	
T0:	No evidence of primary tumor
Tis:	Tumor *in situ* or high-grade dysplasia (HGD)
T1:	Tumor invades lamina propria or submucosa
T2:	Tumor invades muscularis propria
T3:	Tumor invades adventitia
T4a:	Tumor invades resectable structures (e.g., pleura, pericardium, diaphragm)
T4b:	Tumor invades unresectable structures (e.g., aorta, vertebral body, trachea)
Clinical regional lymph nodes (cN)*	
N0:	No regional lymph node metastasis
N1:	Regional lymph node metastasis
Pathologic regional lymph nodes (pN)*	
N0:	No regional lymph node metastasis
N1:	1–2 positive regional lymph nodes
N2:	3–6 positive regional lymph nodes
N3:	≥7 positive regional lymph nodes
Distant metastasis (M)	
M0:	No distant metastasis
M1:	Distant metastasis
Histologic grade (G)	
Gx:	Grade cannot be assessed
G1:	Well differentiated
G2:	Moderately differentiated
G3:	Poorly differentiated
G4:	Undifferentiated

* Defined as periesophageal lymph nodes extending from cervical to celiac nodes.

Adapted from Rice TW *et al.* Cancer of the esophagus and esophagogastric junction: data-driven staging for the seventh edition of the American Joint Committee on Cancer/International Union Against Cancer Cancer Staging Manuals. *Cancer* 2010; **116**(16):3763–73.

Table 15.2 Stage groupings for adenocarcinoma

Stage	T	N	M	G
0	Tis (HGD)	0	0	1
IA	1	0	0	1–2
IB	1	0	0	3
	2	0	0	1–2
IIA	2	0	0	3
IIB	3	0	0	Any
	1–2	1	0	Any
	1–2	2	0	Any
IIIA	1–2	2	0	Any
	3	1	0	Any
	4a	0	0	Any
IIIB	3	2	0	Any
IIIC	4a	1–2	0	Any
	4b	Any	0	Any
	Any	3	0	Any
IV	Any	Any	1	Any

Adapted from Rice TW *et al.* Cancer of the esophagus and esophagogastric junction: data-driven staging for the seventh edition of the American Joint Committee on Cancer/ International Union Against Cancer Cancer Staging Manuals. *Cancer* 2010; **116**(16):3763–73.

Table 15.3 Stage groupings for squamous cell carcinoma

Stage	T	N	M	G	Location
0	Tis (HGD)	0	0	1	Any
IA	1	0	0	1	Any
IB	1	0	0	2–3	Any
	2–3			1	
IIA	2–3	0	0	1	Upper[a], middle[b]
	2–3	0	0	2–3	Lower[c]
IIB	2–3	0	0	2–3	Upper[a], middle[b]
	1–2	1	0	Any	Any
IIIA	1–2	2	0	Any	Any
	3	1	0	Any	Any
	4a	0	0	Any	Any
IIIB	3	2	0	Any	Any
IIIC	4a	1–2	0	Any	Any
	4b	Any	0	Any	Any
	Any	3	0	Any	Any
IV	Any	Any	1	Any	Any

[a]Upper thoracic esophagus, bordered superiorly by the thoracic inlet and inferiorly by the azygos vein. Distance from incisors is 20cm to <25cm; [b]Middle thoracic esophagus bordered superiorly by the azygos vein and inferiorly by the inferior pulmonary veins. Distance from incisors is 25cm to <30cm; [c]Lower thoracic esophagus is bordered superiorly by the inferior pulmonary veins and inferiorly by the stomach. It includes EGJ. Distance from incisors is 30–45cm. Stages 0–IIB are amenable to surgical resection.

Esophagectomy: overview

Key facts

- Numerous approaches to esophagectomy, most of which can be divided into esophagectomy *with thoracotomy* and esophagectomy *without thoracotomy* (Table 15.4).
- There is no survival difference between esophagectomy with thoracotomy and esophagectomy without thoracotomy.[1]
- The stomach is the most common conduit. Conduits include:
  - Stomach: right gastric and right gastric epiploic artery pedicles.
  - Left colon: ascending branch of left colic artery pedicle.
  - Right colon: middle colic artery pedicle.
  - Jejunal free graft: microsurgery implantation of mesenteric arcade.
  - Myocutaneous tube flaps: radial forearm, pectoralis major, TRAM.
- Route of enteric reconstruction is most commonly the posterior mediastinum. If the posterior mediastinum is obliterated, then the conduit can be passed substernally, subcutaneous (anterior to the sternum), or via the right or left pleural space.
- Distal adenocarcinomas are divided by ***Siewert classification*** (Fig. 15.2).
  - *Siewert I:* tumors confined to the distal esophagus, just above the gastroesophageal junction (GEJ), spread to mediastinal and celiac nodes: esophagectomy only required, with negative gastric margins.
  - *Siewert II:* tumors with epicenter in the GEJ, which may invade the proximal gastric cardia: generally esophagectomy only is required, with negative gastric margins. Spread is to abdominal lymph nodes.
  - *Siewert III:* tumors with epicenter within the cardia, and extensive gastric involvement: best treated with total gastrectomy, with possible esophagectomy. Spread is to abdominal lymph nodes.

Choice of esophagectomy (Table 15.4)

- ***Transhiatal:*** portion of esophagus which needs to be blindly mobilized (distal trachea to subcarinal region) should be extrinsically normal as it cannot be directly visualized. Previous thoracotomy, especially if mediastinal pleura opened, mid to upper esophageal transmural cancers, inflammation (e.g., from caustic ingestion injury) are contraindications.

Table 15.4 Esophagectomy

Esophagectomy without thoracotomy	Transhiatal esophagectomy (THE)
Esophagectomy with thoracotomy	1. Ivor Lewis esophagectomy (transthoracic esophagectomy, TTE) 2. McKeown esophagectomy (3-hole) 3. Esophagectomy with three field lymphadenectomy 4. Left thoracoabdominal esophagectomy
Esophagectomy with complete laparoscopic and thorascopic mobilization	Minimally invasive esophagectomy

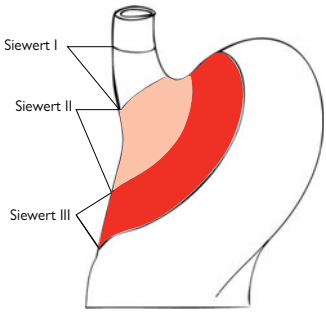


Fig. 15.2 Siewert classification of distal esophageal tumors.

- ***Transthoracic:*** allows mediastinal lymphadenectomy, and accurate resection of tumors in distal tracheal to subcarinal region.
- A VATS approach is limited to patients without N1 or M1 disease.
- Three-field dissection has significant physiologic burden: patients with FEV_1 <1.5L, significant cardiac disease, age >75 years are probably not candidates.

General principles

Esophagectomy should be preceded by esophagoscopy by the operating surgeon to confirm tumor location, extent of tumor and Barrett's mucosa, pyloric narrowing, and duodenal ulceration. Avoiding incomplete tumor resection, anastomotic leaks, and much of the other major morbidity associated with this operation depends on strict adherence to basic general surgical principles:

- This is a major operation conducted in patients who may be elderly, malnourished and immunocompromised, and who often have cardiovascular risk factors:
  - Nutritional support, e.g., NG feeds may be needed preoperatively.
  - Pulmonary physiotherapy including smoking cessation and ambulation may help reduce the high risk of respiratory complications.
  - Normalization of hematocrit and immunological status in patients undergoing preoperative chemotherapy.
  - Careful evaluation of cardiovascular status.
- Gentle handling of tissues.
- Meticulous hemostasis and avoidance of dead space.
- Adequate conduit length to create a tension-free anastomosis.
- Adequate arterial supply.
- Adequate venous drainage (do not dissect out more distal esophagus more than necessary to mobilize it).
- Gastric decompression, including pyloroplasty or pyloromyotomy in patients where gastric tube reconstruction is chosen, and placement of an NG tube to reduce risk of vomiting.

Reference

1. Chang AC *et al.* Outcomes after transhiatal and transthoracic esophagectomy for cancer. *Ann Thorac Surg* 2008; **85**:424–9.

Transhiatal esophagectomy

Key facts

Esophagectomy without thoracotomy (most suitable for lesions when esophagus from distal trachea to subcarinal region is extrinsically normal as this is blindly mobilized). It is performed in three phases:

- *Abdominal phase* (Fig. 15.3, top left):
  - Performed through an upper midline laparotomy incision.
  - Divide short gastric, left gastroepiploic, and left gastric arteries.
  - Kocher maneuver followed by a pyloromyotomy.
  - Esophageal hiatus, and the distal esophagus mobilized by sharp and blunt dissection.
- *Cervical phase:*
  - Standard left neck incision anterior to sternocleidomastoid.
  - Middle thyroid vein and inferior thyroid artery are divided.
  - The superior esophagus is mobilized down to the carina.
- *Mediastinal phase* (Fig.15.3, bottom left):
  - Mobilize posterior mediastinum—sharp and blunt dissection.
  - The cervical esophagus is divided and the proximal stomach is divided with a 4–5cm margin from the tumor.
  - Gastric conduit delivered via posterior mediastinum to neck.
  - A feeding jejunostomy is placed and the abdomen closed.
  - A cervico-esophagogastric anastomosis is performed end-to-side either stapled or hand sewn, after a NG tube is positioned.

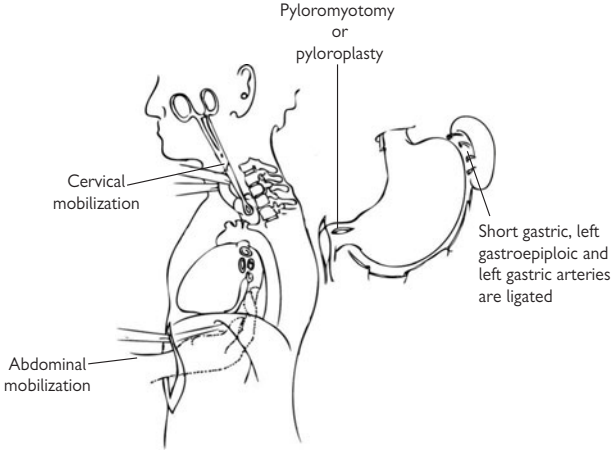


Fig. 15.3 Transhiatal esophagectomy.

Set-up

- Single lumen ET tube, NG tube, arterial line, Foley catheter.
- CVP line optional.
- Supine with left arm tucked, right arm extended on arm board.
- Roll under neck improves cervical exposure.
- Prep neck, chest, and abdomen to groin as continuous field.

Operative procedure

Gastric mobilization

- Upper midline laparotomy, extending over xiphoid to lower sternum.
- Retractor with bladder blade to lift xiphoid, malleable blade on right to retract left hepatic lobe exposing esophageal hiatus and EGJ.
- Gastric mobilization preserving left gastroepiploic arcade.
- Left gastric vessels exposed and ligated individually with 3/0 silk (or staple/Harmonic scalpel) via lesser sac or anteriorly via divided gastrohepatic ligament.
- Kocher maneuver to straighten duodenum (peritoneum incised to right of duodenum which is reflected to left to expose structures behind).
- Pyloromyotomy (15 blade or electrocautery on low longitudinal incision muscularis down to mucosa), cover with omental patch.

Esophageal mobilization

- Performed from below through esophageal hiatus.
- Widen hiatus anteriorly after ligating crossing phrenic vein, or by dividing left crus (more risk breaching left pleural space).
- Blunt mobilization of esophagus into upper mediastinal space (NG tube helps stay on esophagus).

Cervical phase

- Modified left collar incision.
- Anterior border sternomastoid is retracted laterally, prevertebral space accessed medial to carotid sheath.
- Distal cervical esophagus encircled, bimanual mobilization to complete intrathoracic dissection.
- Laryngeal nerve injury a risk.

Creation gastric tube and cervical anastomosis

- Based on greater curvature stomach, 4–5cm diameter using serial firings of 7.5cm GIA stapler, oversew staple line to prevent fistula/leaks.
- Deliver esophagus and gastric cardia from chest—either pull esophagus through neck, or divide distal cervical esophagus and deliver tumor through abdomen.
- Attaching a 32F chest tube to stomach tip helps prevent twisting.
- Hand-sewn anastomosis (4/0 silk two-layer, inverting).
- Direct NG tube across anastomosis under direct vision before completing.
- Replace anastomosis in midline, prevertebral space.

Closure

- Irrigate and close cervical incision in layers.
- Place adjuvant jejunostomy, abdominal incision closed in layers.
- Abdominal and cervical drains.

Esophagectomy with thoracotomy

Key facts

- Four main types of esophagectomy involve thoracotomy (or thoracoscopic approaches), improving exposure of the thoracic esophagus, with anastomoses in either the thorax or the neck.
- ***Transthoracic esophagectomy (Ivor Lewis)*** allows mediastinal lymphadenectomy, and accurate resection of tumors in the distal tracheal to subcarinal region (Fig. 15.4).
- ***A three-hole (McKeown)*** approach combines the excellent exposure of the Ivor Lewis with the advantages of a cervical anastomosis, as well as the option of performing lymph node dissection in three fields (Fig. 15.4).
- ***Three-field dissection*** has significant physiologic burden: patients with FEV_1 <1.5L, significant cardiac disease, age >75 years are probably not candidates.
- ***A left thoracoabdominal approach*** employs a single incision in patients with tumors at the gastroesophageal junction (Fig. 15.4).

Ivor Lewis (transthoracic) esophagectomy

Rationale

- Esophagectomy with right thoracotomy named after Welsh surgeon, Mr Ivor Lewis, designed to provide excellent exposure of the intrathoracic mid to upper thoracic esophagus.
- It also allows the surgeon to perform a regional lymph node dissection under direct vision.
- The main drawbacks compared to a transhiatal esophagectomy are the additional morbidity of a thoracotomy incision, and an intrathoracic anastomosis (leaks from a cervical anastomosis are easier to treat and less likely to result in respiratory and systemic complications).

Technique

- Performed through two incisions in two phases.
- ***Abdominal phase:***
  - Performed through an *upper midline laparotomy* incision.
  - Steps similar to transhiatal esophagectomy.
  - Abdomen is closed after mobilization of the stomach conduit.
- ***Thoracic phase:***
  - Performed through a *right posterolateral thoracotomy* after repositioning the patient in the right lateral position.
  - The esophagus is mobilized and the azygos vein is divided.
  - The thoracic duct is mass ligated prophylactically near the hiatus.
  - The stomach conduit is brought into the chest. The esophagus is divided at the level of the azygos vein, and the proximal stomach is divided as mentioned on 📖 p767). The thoracic esophagogastric anastomosis (TEGA) is performed either end-to-side stapled or hand sewn, or stapled end-to-end, after a NG tube is positioned.

McKeown or three-hole esophagectomy

- Three incisions; modified left collar cervical incision, upper midline laparotomy, and right posterolateral thoracotomy.
- This combines the advantages of exposure of an Ivor Lewis esophagectomy, with the advantages of a cervical anastomosis.
- No part of the dissection is performed blindly.
- The abdominal and thoracic phases are performed similarly to Ivor Lewis esophagectomy, although the thoracic phase is performed first.
- A cervico-esophagogastric anastomosis is performed in a similar way to the transhiatal esophagectomy.

Left thoracoabdominal approach

- This employs a single incision extending from the left 5th or 6th intercostal space in the anterior axillary line, across the costal margin to the abdomen.
- It provides best exposure of the lower 1/3 of the esophagus, so is used in patients with tumors at the gastroesophageal junction.
- The disadvantages are that the amount of proximal esophagus that can be visualized and therefore mobilized is limited (the aortic arch is in the way), and division of the costal margin contributes to post-thoracotomy pain and respiratory dysfunction.

Minimally invasive esophagectomy

Many different approaches, including robotic, but common approach is similar to Ivor Lewis, with a thoracic esophagogastric anastomosis, however, the abdominal and thoracic phases are performed laparoscopically and thorascopically respectively.

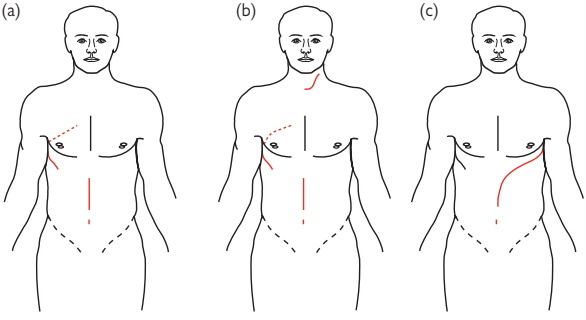


Fig. 15.4 Esophagectomy approaches. (a) Ivor Lewis. (b) McKeown or three-hole. (c) Left thoracoabdominal.

Lymphadenectomy

Key facts

- The optimal extent of lymph node dissection in esophageal cancer is controversial.
- The esophagus has an extensive lymphatic drainage, which is divided into three zones (fields): abdominal, intrathoracic, and cervical.
- Lymph node resection strategies at the time of esophagectomy range from standard regional and one-field lymphadenectomy, to radical approaches involving two- or three-field lymphadenectomy.

Anatomy of lymphatic drainage

- The esophagus has a dense network of mucosal and submucosal lymphatics: hence tumors as early as stage II have access to draining lymphatics and therefore regional spread.
- Tumors of the cervical and middle esophagus generally drain to upper mediastinal lymph nodes.
- The middle and lower 1/3 of the esophagus drains to lymph nodes along the lesser curvature of the stomach, celiac axis nodes, cardiac and upper and lower paraesophageal nodes.
- Celiac axis and lesser curvature nodes are involved in 60% of tumors of the lower 1/3 of the esophagus: celiac nodal involvement represents distant spread and precludes surgical cure.
- The *Siewart classification* of distal esophageal tumors (📖 p765) differentiates between type I lesions which tend to drain to mediastinal and celiac lymph nodes, as opposed to the more distal type II and III lesions which drain to celiac, splenic, and para-aortic nodes.
- Squamous cell cancers are characterized by extensive local and regional involvement with skip metastases even in relatively early stage cancers, which underlies the rationale for more extensive lymphadenectomy.
- Early-stage adenocarcinomas on the other hand tend not to have skip lesions, with spread generally limited to locoregional nodes; consequently the benefits of radical lymphadenectomy are more controversial.

Results of lymphadenectomy approaches

- There are no large-scale randomized studies to support radical lymphadenectomy over more limited lymph node resection with chemoradiotherapy.
- Radical procedures are more complex, with higher incidence of major postoperative complications: there is still around a 20% locoregional recurrence rate.
- Operative mortality is comparable to standard esophagectomy, but comparative survival data is mixed with some evidence suggesting enhanced survival with en bloc esophagectomy in early and later stage cancers, compared to standard transhiatal technique, but other studies finding no difference.
- Variation in surgical technique, and discrepancies in preoperative staging, and intraoperative lymph node sampling and resection may account for some of these differences.

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Chemotherapy and radiotherapy

Key facts

- Treatment options include primary surgery, neoadjuvant chemoradiation followed by surgery, postoperative adjuvant chemoradiotherapy, definitive or primary chemoradiation, palliative chemotherapy ± radiotherapy.
- Neoadjuvant chemoradiotherapy followed by surgery probably offers best survival in patients with locally advanced resectable tumor: cure rate is <50% due to a combination of inadequate pathologic response and distant metastases:
  - Stage 0, I, and IIa should be offered primary surgery.
  - Stage IIB and III tumors should be offered trimodality therapy.
  - Stage IV tumors are treated with primary chemoradiotherapy.
- Nutritional status should be optimized before treatment, possibly requiring either NG or jejunal tube feeds.
- PEG should be avoided, if possible, if the stomach is planned as a conduit. PEG may injure the right gastroepiploic artery which is a necessary vascular pedicle for a gastric conduit.

Definitions

- ***Neoadjuvant therapy*** is treatment (chemotherapy and/or radiotherapy) given *before* surgery. Also called ***induction*** or ***preoperative therapy***.
- ***Adjuvant therapy*** is treatment (chemotherapy and/or radiotherapy) given *after* surgery.
- ***Chemoradiotherapy*** is chemotherapy and radiotherapy given simultaneously.
- ***Primary therapy*** is therapy given as the first line of treatment with intent to cure (as opposed to palliative therapy).
- ***Trimodality therapy*** comprises neoadjuvant therapy, surgery, and then adjuvant therapy.
- ***Clinical stage*** is the tumor stage determined by multiple imaging including laparoscopic assessment, but without pathologic examination of the resected tumor.
- ***Pathologic stage*** is the tumor stage determined by inspection of the resected specimen.

Treatment options

Neoadjuvant chemoradiation

- Rationale for neoadjuvant chemoradiation:
  - Chemotherapy and radiation offer synergistic effect on the locoregional level.
  - Chemotherapy treats *micrometastatic* disease.
  - Chemotherapy is better delivered to primary tumor when the blood supply is intact.
  - Downstaging by chemoradiation may increase resectability.
- Two major studies demonstrate survival advantage for neoadjuvant chemoradiation followed by surgery compared to surgery alone for stage IIB-III esophageal cancer.
  - Urba *et al.*[1] Regimen included 5-FU/cisplatin/vinblastine/concurrent 45Gy. Significant decrease in local recurrence and trend in survival at 3 years.

- Walsh *et al.*[2] 5-FU/cisplatin/concurrent 40Gy. Significant increase in survival at 3 years.

- Two major meta-analyses demonstrate advantage for neoadjuvant chemoradiation followed by surgery compared to surgery alone for stage IIB–III esophageal cancer.
  - Urschel *et al.*[3] showed improved 3-year survival, reduced locoregional tumor recurrence, higher rate of complete (RO) resection, overall complete pathologic response rate of 21%.
  - Gebski V *et al.*[4] showed improved 2-year survival, survival benefit for both adenocarcinoma and SCC. There was no survival benefit for sequential chemoradiotherapy for SCC.
- Common neoadjuvant regimen: cisplatin or carboplatin and 5-FU or taxol with concurrent 45Gy or 50.8Gy external beam radiation, 3–8 weeks of rest, followed by surgery.

Adjuvant therapy

Patients with R1 or R2 resection or residual postoperative N1 disease should be considered for adjuvant chemotherapy ± radiation, though there is no conclusive data that supports this regimen.

Primary chemoradiation

- Should be considered in patients who from a cardiopulmonary or anatomic standpoint are poor surgical candidates, or who refuse surgery, and for stage IV tumors.
- Two studies suggest role for definitive chemoradiation in patients who are otherwise surgical candidates.
  - Stahl M *et al.*[5] randomized 172 patients with locally advanced (T3,4, N0–1, M0) SCC of the esophagus who received induction (5-FU/etoposide/cisplatin) chemotherapy followed by chemoradiotherapy (40Gy to surgery, or continued chemoradiotherapy (65Gy)). No 2-year survival difference. 31% to 24% survival advantage for surgery, not statistically significant. Survival curves widened after 3 years.
  - Bedenne *et al.*[6] randomized 259 patients with locally advanced (T3, N0–1, M0) cancer (88% SCC) of the esophagus. Received induction (5-FU/cisplatin) chemotherapy followed by concomitant radiotherapy (46Gy) conventional or split course. Patients with tumor response were then randomized to surgery or continued chemoradiotherapy. No difference in 2-year survival. Patients undergoing surgery had lower local recurrence (34% vs. 43%) and less need for additional palliation (24% vs. 43%).

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Postoperative care

Key facts

- 40–50% of esophagectomy patients have some sort of postoperative complication (📖 pp776–777).
- Postoperative pathways and protocols can be helpful.

Immediate perioperative and postoperative care

- Intraoperative pressors should be avoided, and adequate intravascular volume should be maintained.
- Transfusion of blood products in the setting of cancer may increase the incidence of recurrence. Transfusion may be considered if the hemoglobin level drops below 8g/dL.

Pulmonary toilet

- Aggressive pulmonary toilet and early ambulation to prevent pneumonia.
- Patients should ambulate 4×/day.
- Head of bed at 45° to reduce aspiration risk.

Nutrition

- An intrathoracic anastomosis should be examined with a thin barium esophagram before instituting an oral diet (day 4–5). Oral diet can be instituted in a stepwise fashion for cervical anastomosis.
- Diet should start with sips of liquids, to clear liquids, full liquids, and finally soft, ground diet.
- Tube feedings via the jejunostomy tube can be initiated beginning postoperative day 3 and advanced to goal within 48–72h. Rate of tube feeds may need to be adjusted or held if development of an ileus.

Deep venous thrombosis prophylaxis

- Prophylaxis dose of LMWH is used (e.g., 5000U heparin sc bid or tid) until the patient is discharged from the hospital.
- In addition, sequential compression devices are worn while the patient is in bed or sitting in a chair.
- Early ambulation also assists in DVT prevention.

Gastrointestinal prophylaxis

- Although outside of a vagal-sparing esophagectomy, the vagi are removed and acid production by the conduit is diminished; however, IV and then PO PPI are used to prevent gastric secretions of the conduit.
- The NG tube is generally removed postoperative day 3–7.
- IV antiemetics are used if necessary, though incidence of nausea after esophagectomy is no higher than other general surgeries.

Palliation

- Attempts at palliation are generally directed to alleviating or minimizing dysphagia.
- Photodynamic therapy (PDT):
  - The patient is injected with a 2mg/kg photosensitizer such as sodium porfimer (Photofrin®)
  - After 48h the tumor is exposed to 300–400J/cm of 630nm light.
  - <5% perforation rate.
  - Skin injury from light exposure is common.
- Brachytherapy:
  - Catheter-based administration of iridium-192.
  - Risk of fistula formation.
- Esophageal stents:
  - Endoscopy-directed plastic or self-expanding metal stents (SEMS).
  - Plastic stents are easily removed, but frequently migrate.
  - Nickel/titanium (nitinol) stents are common. Covered stents (silicone or polyurethane) are beneficial in temporarily sealing trachealesophageal fistulas. 95% effective in alleviating dysphagia.

Complications and outcomes

Perioperative

Pneumothorax

During the gastric mobilization and distal mediastinal dissection, both mediastinal pleural spaces are inspected to determine pleural disruption. Chest tubes are placed if identified. Prior to extubation in the operating room, an intraoperative CXR is obtained to determine pneumothorax of pleural effusion, and a chest tube is placed if necessary.

Hemorrhage

Although arterial branches from the aorta to the thoracic esophagus are avulsed during THO, significant bleeding is rarely an issue as such vessels usually undergo spasm. If the azygos vein is disrupted during dissection during a THO, simple packing usually stops bleeding, as the azygos vein is part of a low-pressure system.

Tracheal injury

Tracheal injury during esophagectomy should be repaired once identified. Usually the injury involves the membranous wall of the trachea, and is identified by release of air from the wound. If this occurs during the cervical dissection, a partial sternal split can be performed to provide exposure for repair, or the patient should be positioned for a right thoracotomy if the injury is more distal on the trachea.

Postoperative

Recurrent laryngeal nerve injury

- Rare complication after THE.
- Hoarseness occurs in 5% of patients, permanent chordal paralysis in <1%.
- Injury to the recurrent laryngeal nerve can be avoided by keeping dissection close to the surface of the esophagus, avoiding the tracheoesophageal groove, and not using metal to retract the trachea medially.

Supraventricular tachyarrhythmias

- AF occurs in up to 30% of thoracic surgical patients.
- AF prophylaxis can be utilized, including magnesium and β-blockade (see 🕮 pp262–263).

Chylothorax

- Rare complication. Higher after THE (1%).
- Chylothorax should be dealt with immediately with diet modification, and possible surgical intervention.
- During esophagectomy with thoracotomy, the thoracic duct can be ligated prophylactically.

Pleural effusions

- Pleural effusions should be drained with small- or regular-bore chest tubes.
- Late development of a pleural effusion with an ipsilateral intrathoracic anastomosis should raise suspicions of an anastomotic leak.

Anastomotic leaks

- *Cervical esophagogastric anastomosis* (4–12%) have higher incidence of anastomotic leaks than intrathoracic anastomosis.
- Once an anastomotic leak develops, 50% of those anastomoses go on to develop strictures.
- *Cervical leaks* should undergo early dilation.
- 90% of cervical esophagogastric anastomosis leaks occur within first 10 postoperative days.
- Risk factors for developing CEGA leaks include higher number of preoperative comorbidities, advanced pathologic stage, postoperative arrhythmia, increased number of prior esophagogastric surgeries, and active smoking history.
- Previous chemoradiation does not seem to be a risk factor for anastomotic leak.
- CEGA leaks often present as foul discharge from the wound, low-grade temps, wound erythema, leukocytosis, halitosis.
- Cervical leaks are treated by bedside opening of the wound and twice daily packing. The patient finds a position where they can drink and eventually eat without leaking. The fistula usually heals within 2 weeks. Early dilation stretches an edematous anastomosis and promotes antegrade movement of swallowed food.
- *Intrathoracic leaks* present as fever, tachycardia, tachypnea, new ipsilateral pleural effusion, and can be treated by adequate drainage with chest tube(s) and an endoscopic-placed covered stent across the anastomosis.

Outcomes

- Esophagectomy with and without thoracotomy have similar perioperative and long-term outcomes.
- Univariate analysis demonstrates that THE has reduced overall complications (P = 0.01) including pulmonary and GI complications. However, after controlling for confounding factors, multivariate analysis, THE and TTE have similar overall complication rates, though TTE has slightly higher gastrointestinal and systemic complications.[1]
- In regards to cancer specific survival, studies demonstrate that the number of lymph nodes obtained as well as the number of surgical fields dissected, may improve overall cancer specific survival. However, adjusted for patient characteristics (hazard ratio (HR) 0.91) and patient and provider characteristics (HR 0.95) THE and TTE have no difference in oncologic survival.[2]
- Cancer specific survival after neoadjuvant chemoradiotherapy followed by surgery is largely dependent on the pathologic stage of the tumor. Complete pathologic (yp) response demonstrates a 2- and 5-year survival of 80% and 58% respectively. A yp stage of I–IV combined yields a 2- and 5-year survival of 45% and 22% respectively.[3]

References

1. Connors RC *et al.* Comparing outcomes after transthoracic and transhiatal esophagectomy: a 5-year prospective cohort of 17,395 patients. *J Am Coll Surg* 2007; **205**:735–40.
2. Chang AC *et al.* Outcomes after transhiatal and transthoracic esophagectomy for cancer. *Ann Thorac Surg* 2008; **85**:424–9.
3. Orringer MB *et al.* Two thousand transhiatal esophagectomies: changing trends, lessons learned. *Ann Surg* 2007; **246**:363–72.

Gastroesophageal reflux disease

Key facts

- GERD is the reflux of acid or bile from the stomach into the esophagus caused by dysfunction of the lower esophageal sphincter (LES); or changes in the type or volume of refluxate, the mucosal barrier, or esophageal clearance.
- GERD can result in substernal chest pain, but the primary rationale for treating it remains the risk of Barrett's esophagus (📖 p780) and potential for malignant transformation.
- If antacids, and dietary modification fails, laparoscopic Nissen fundoplication and wrap is the procedure of choice.

Pathology

- The normal '*anti-reflux barrier*' is made up of the intrinsic pressure of the LES, its intrabdominal location, the crural diaphragm, the acute angle of His between the esophagus and stomach.
- The normal LES has a resting amplitude of 12–20mmHg and length of 3–5cm. A dysfunctional LES has sphincter pressure <6mmHg and/or overall sphincter length <2cm. Causes of LES dysfunction include:
  - Cephalad displacement of LOS, e.g., sliding hiatal hernia.
  - Obesity and increased intra-abdominal pressure.
  - Foods and drugs, including nicotine, nitrates, benzodiazepines, mint, caffeine, chocolate, tomato-based products, spiced foods, alcohol.
- Changes in refluxate (increased acid and pepsin, or bile salts), reduced esophageal clearance (type II hiatus hernia), pregnancy, previous gastric surgery, scleroderma, and diabetes can contribute to GERD.

Presentation and investigation

- May be asymptomatic. Atypical substernal chest pain (heartburn), belching, halitosis, cough, wheeze, dysphagia, reflux esophagitis, stricture.
- *Barium esophagram* can identify a sliding hiatal hernia, after reflux of contrast material and presence of strictures.
- *Endoscopy* (📖 p658) determines the presence of esophagitis, and biopsy abnormal mucosa, and identifying malignancy or Barrett's metaplasia.
- *Manometry* is used to determine the location of the LES or high-pressure zone (HPZ) and document abnormal LES tone.

24h pH monitoring

- The patient holds H2 receptor blockers or proton pump inhibitors (PPIs) for 48–72h prior to the exam.
- The pH probe is placed 5cm above the HPZ. Six components are recorded to determine a composite DeMeester score:
  - Total percentage of time pH <4.
  - Percent upright and supine time pH <4.
  - Number of total reflux episodes.
  - Number of reflux episodes that are ≥5min in duration.
  - Longest reflux episode duration in minutes.
- Abnormal esophageal acid exposure is diagnosed if composite DeMeester score >14.7, or the percent total time pH <4 is >4%.

Management

Medical management includes diet modification (eliminating precipitants listed on 📖 p776), antacids, i.e. H_2 receptor blockers, e.g. cimetidine 150mg twice daily, or proton pump inhibitors (PPI), e.g. omeprazole 20mg daily.

Indications for surgery

Diagnosis of GERD as defined by barium esophagram, manometry and 24h pH monitoring, and:

- Failed medical management, but responsive to PPI therapy.
- Presence of esophagitis.
- Barrett's metaplasia.
- Recurrent aspiration.

Laparoscopic Nissen fundoplication

- 360° wrap.
- Should be avoided in patients with a concurrent motility disorder as complete wrap may cause dysphagia.
- Nissen fundoplication can be done open or laparoscopically. Unless a multiple redo fundoplication, most cases can be done laparoscopically.
- Five ports may be used, with one port used for specifically for liver retraction.
- The gastrohepatic ligament (pars lucida) is divided, as well as the phrenoesophageal ligament exposing the esophageal hiatus, and the right and left crus.
- The short gastric arteries are divided.
- Opening the hiatus, the distal mediastinum is dissected. To avoid tension and a slipped wrap, at least 2.5cm of esophagus must be within the abdomen without tension, otherwise an esophageal lengthening procedure (Collis gastroplasty) is necessary.
- The crural is closed to the approximation of the esophagus with a 56F Maloney dilator.
- Cural closure can be reinforced with biodegradable reinforcement (xeno- or allograft patches).
- Synthetic mesh should not be used to avoid extrinsic stricturing and perforation.
- The fundus of the wrap is slid posteriorly and around the esophagus to form a 360° wrap, and secured with interrupted sutures.

Postoperative care

- The patient is started on clear liquids and advanced to soft solids.
- Carbonated beverages and thick foods like bread and steak are avoided.
- Complications include:
  - Esophageal perforation and other perforated viscus, splenic injury, liver injury or injury during initiating pneumoperitoneum.
  - Slipped Nissen, or herniation of the wrap into the mediastinum may occur if the abdominal esophagus is under tension.
- Can be repaired laparoscopically if found early.

Barrett's metaplasia

Key facts

- Barrett's metaplasia describes the situation where *the squamocolumnar junction is cephalad to the gastroesophageal junction* (GEJ). Initially described by Australian thoracic surgeon Norman 'Patsy' Barrett in 1950.
- It is an adaptive response to esophageal mucosal injury from gastric acid or bile reflux. It is found in 6–12% of patients with GERD.
- Its clinical relevance is the increased risk of malignant transformation (around 1% of patients per year with Barrett's metaplasia progress to adenocarcinoma of the esophagus).

Pathology

- Presence of columnar mucosa (intestinal metaplasia) with goblet cells, with progressive disorganization of cells and glands, above the GEJ:
  - No dysplasia.
  - Low-grade dysplasia (LGD)
  - High-grade dysplasia (HGD).
  - Adenocarcinoma.
- 5–10% of patients per year will progress to dysplasia.
- 15–25% of patients have concurrent LGD.
- 5–10% of patients have concurrent HGD.
- Risk for adenocarcinoma is increased 30–125-fold.
- 0.5–1% per year will progress to adenocarcinoma.
- Occult cancer is found in 25–73% of esophagectomy specimen for HGD.
- Primary Barrett's results from esophageal mucosal injury from gastric acid or bile reflux. Other causes include salivary nitrates, tobacco, oral flora, obesity, increased insulin-like growth factors, bacterial colonization, *Helicobacter pylori* eradication, hypergastrinemia, and chronic use of proton pump inhibitors.

Management

Surveillance

- Barrett's metaplasia without dysplasia: endoscopy every 2–3 years with four-quadrant biopsies at 2cm intervals.
- LGD: endoscopy and four-quadrant biopsies at 2cm intervals ever 6 months for 1 year, then annually if still LGD.
- HGD: confirm diagnosis with repeat endoscopy and two separate pathologists. If confirmed then procedural intervention:
  - Flat and unifocal HGD may be treated with endoscopic techniques
  - Multifocal HGD, HGD with dysplasia-associated lesions or masses (DALM) and long segments of dysplasia are best treated by esophagectomy.

Endoscopic therapy

- *PDT*: injection of photosensitizing agent followed by treatment with 630nm laser causes oxygen-radical dependent tissue necrosis. Complications include 30% esophageal stricture, 10% sunburn odynophagia. Key limitations:

- Depth of penetration limited to submucosa.
- Invasive cancers are inadequately treated.

- *Radiofrequency ablation*: ablates tumor to the level of the muscularis mucosae. Does not provide intact histology architecture and therefore provides no information on deep or lateral margins.
- *Endoscopic mucosal resection*: injection of air or saline in submucosa beneath lesion, and then endoscopic snaring, or suctioning. Provides large specimen with intact histology architecture to assess deep and lateral margins.

Surgery

Esophagectomy is indicated for HGD (📖 pp764–765).

Esophageal strictures and webs

Key facts

- Esophageal rings and webs are the most common structural abnormalities in the esophagus (found in around 10% esophagrams).
- Strictures should be biopsied to rule out malignancy (📖 p658).
- Esophageal webs and most strictures respond well to dilatation and rarely need surgery.

Benign esophageal strictures

Etiology

See Table 15.5.

Pathology

Abnormal narrowing of the esophageal wall. Most commonly as a result of chronic inflammation leading to fibrosis. Peptic strictures represent 75% of benign esophageal stricture.

Presentation and diagnosis

- Symptoms may include dysphagia to solids and then liquids, odynophagia, hemoptysis, impaction of food.
- *Barium esophagram* can identify stricture location. Important to evaluate esophagram at multiple images, in order to distinguish a stricture from a transient peristaltic indentation.
- *Endoscopy* is important for visual characterization and biopsy. Never push a food impaction forward, as there may be a stricture behind the impaction, and esophageal perforation may result.

Management

- *Blind dilation* using *Maloney dilators* can be performed for cervical anastomotic strictures after esophagectomy.
- *Endoscopic dilation* with either wire-guided Savary dilators or pneumatic balloon dilation.
- Refractory symptomatic strictures may require *esophagectomy* (see 📖 p764).

Esophageal rings and webs

Pathology

- Thought to be congenital, or due to inflammation, postcricoid webs associated with Pulmmer–Vinson syndrome.
- The terms rings and webs are often used interchangeably. An esophageal ring is a concentric smooth, thin (3–5mm) extension of normal esophageal tissue including the three anatomic layers (mucosa, submucosa, and muscle) found anywhere along the esophagus, but usually distal. Three types:
  - A ring: uncommon. Several cm proximal to squamocolumnar junction (SQJ). Not a fixed structure, may not appear in esophagrams.
  - B (Schatzki) ring: most common, composed of mucosa and submucosa only, usually located at the SQJ marking proximal margin of a hiatal hernia (📖 p788).
  - C ring: rare finding, caused by indentation of diaphragmatic crura.
  - Webs: thin, eccentric, extension of esophagus mucosa and submucosa occurring anywhere along esophagus but usually postcricoid.

Presentation

- Most asymptomatic. Almost all patients with lumen <13mm have dysphagia, worse for solids than liquids (in contrast with motility disorders). Location of ring or web often determines location of sensation of dysphagia, or discomfort.
- Weight loss and anemia suggest underlying malignancy.

Investigation

- Barium swallow demonstrates intraluminal obstruction, competency of the LES and presence of GOR, and is more sensitive than OGD.
- OGD is useful for biopsy if malignancy is suspected.

Management

- If symptoms not managed by conservative measures such as modifying diet, esophageal dilatation can be carried out using Maloney or Savary dilators, with a high success rate and minimal recurrence.
- Surgery is rarely indicated.

Table 15.5 Etiology of benign esophageal strictures

Congenital
Webs: Perforate Imperforate
Acquired
Peptic
Webs: Plummer–Vinson syndrome Schatzki's ring
Autoimmune: SLE Scleroderma Polyarteritis nodosa Dermatomyositis
Caustic injury
Chronic foreign body reaction
Crohn's disease
Infectious: Tuberculosis Syphilis
Radiation effect
Postoperative anastomotic
Drug-induced
Idiopathic

Esophageal diverticula

Key facts

- These are blind pouches lined by epithelium arising from the esophagus: although they are usually acquired as a result of motility disorders (e.g., *Zenkers*), they may occasionally be congenital, or due to local inflammatory disease and compression.
- They may be asymptomatic, or present with dysphagia, odynophagia, regurgitation, and aspiration.
- Diverticula are excised open or endoscopically: there is no role for medical management.

Pathology

- Esophageal diverticula are classified as pulsion or traction diverticulae:
  - *Pulsion (or pseudo) diverticulae* consist of only mucosal layers herniating through a weak point as a result of increased transmural pressure gradient, and include Zenker's and epiphrenic diverticula.
  - *Traction (or true) diverticula* consist of all three muscle layers, and are caused by inflammation or fibrosis in adjacent lymph nodes.

Zenker's diverticulum

Pathology

- This is a propulsion diverticulum involving only the mucosa layer.
- It protrudes posteriorly from *Killian's triangle*, an area of anatomic weakness bounded by the inferior constrictors and cricopharyngeus muscle or upper esophageal sphincter (UES)
- Underlying mechanism is non-coordinated deglutition with premature contraction and incomplete relaxation of UES.
- The neck lies above cricopharynx lading to spontaneous emptying and regurgitation into mouth, and aspiration, and compression of pharynx.

Presentation and diagnosis

May be asymptomatic. Presents in 5th–8th decades with dysphagia, *regurgitation of fresh food*, aspiration, halitosis, hoarseness, weight loss, asthma.

- Barium esophagram is gold standard.
- Endoscopy.

Management

- There is no medical therapy: all patients are candidates for surgery.
- Diverticulectomy and myotomy are best formed electively while diverticulum is still small.
- Myotomy alone can be performed for small (<2cm) diverticulum.
- *Open approach*: the cervical esophagus is reached via a left neck incision, anterior to sternocleidomastoid, and anterior to the carotid towards the cervical spine. A long myotomy is performed from the inferior constrictors superiorly, dividing the cricopharyngeus muscle, extending inferiorly dividing both muscle layers of the esophagus towards the clavicle. The diverticulum is divided with a stapler.
- *Endoscopic approach*: intraoperative trans-oral stapling of the esophageal wall and diverticulum. A Collard–Weerda diverticuloscope

is placed simultaneously in the diverticulum and the esophagus and a stapler is used to simultaneously divide the anterior wall of the diverticulum, cricopharyngeus.
- 4% conversion rate, 4% reoperation rate. 1.5% mortality, recurrent laryngeal nerve palsy in 2%, fistula in 2%.

Epiphrenic diverticulum

Pathology

Rare pulsion-type diverticula, only involving the mucosa and submucosa, found in the distal 10cm of the esophagus, found predominantly in elderly people, and associated with achalasia and GERD.

Presentation and investigation

- Symptoms variable: dysphagia, odynophagia, regurgitation, aspiration.
- Barium esophagram is the gold standard for diagnosis.
- Chest CT aids in characterization of anatomy.

Surgery

Performed via a left thoracotomy, even if diverticulum extends into the right chest. This is a motility disorder, therefore a long myotomy is performed, extending onto the wall of the stomach. The diverticulum is divided with a stapler. To prevent reflux, a modified Belsey antireflux procedure is performed. Surgical outcomes 2.8% mortality and 5.7% leak rate.[1] Key is to perform a myotomy which limits risk of leak after diverticulum resection.

Other acquired diverticula

Traction diverticula

- These are mid-esophageal diverticula that involve all layers of the esophagus wall, and are not related to disorder in motility.
- They occur as a result of chronic traction or pulling from chronic mediastinal inflammation and fibrosis, e.g., histoplasmosis, TB.
- Symptoms are rare because their location and size, but may include dysphagia, odynophagia (due to stricture/compression), and bleeding.

Management

- Local excision or stapled resection of the diverticulum and adjacent inflammatory mass, best approached via a right thoracotomy.
- Extensive myotomy is not required.

Congenital diverticula

- These rare diverticula are true diverticula containing esophageal mucosa, submucosa, and muscular walls: embryology is uncertain.
- They usually present in late infancy with progressive dysphagia, recurrent respiratory infections, or in the case of high diverticula regurgitation, cough with feeding, excess salivation, and even cyanosis.
- They are diagnosed with a contrast esophagram and endoscopy
- Surgical excision is via a single-stage transverse diverticulectomy.

Reference

1. Varghese TK *et al.* Surgical treatment of epiphrenic diverticula: a 30-year experience. *Ann Thorac Surg* 2007; **84**(6):1801–9.

Motility disorders

Key facts

Esophageal motility disorders include problems of hypomotility (e.g., achalasia) and hypermotiity (diffuse esophageal spasm, nutcracker esophagus and hypertensive LES).

Physiology of swallowing

- Esophageal function is coordinated by sympathetic innervation (from the ***cervical and thoracic chains and celiac plexus***), parasympathetic innervation from the ***vagus which forms Auerbach's plexus*** around the circular muscular layer) and ***Meissner's submucosal plexus***.
- Peristalsis is coordinated under central control through the vagus nerve by Meissner's and Auerbach's plexus:
  - Primary peristalsis is a normal propulsive wave in response to the stimulus of voluntary deglutition.
  - Secondary peristalsis is a normal wave without voluntary deglutition.
  - Tertiary waves—abnormal low-amplitude simultaneous contractions.
- For effective swallow peristalsis is coordinated with contraction and relaxation of the ***upper (UES) and lower (LES) esophageal sphincters***.
  - The UES is composed of cricopharyngeus and inferior pharyngeal constrictors at C5–C6, remains contracted between swallows due to stimulation by ***CN IX and XI***, with resting tone up to 100mmHg. Swallowing involves ***inhibition*** of IX and XI, UES opens then closes.
  - The LES is specialized arrangement of esophageal muscle 3–4cm above GEJ, has a high-pressure zone with resting tone 15–25mmHg. Influenced by neural, hormonal and drugs: relaxes during swallow and closes with passage of food bolus through LES.

Achalasia

Pathology

- Achalasia is a rare (1:100 000) disorder characterized by ***loss of peristalsis*** in the distal esophagus and ***failure of the LES to relax***. Some patients have abnormal vigorous tertiary peristalsis:
  - *Primary achalasia* is an idiopathic degenerative nerve process associated with loss of ganglion cells in the myenteric plexus, and reduction of vagal fibres in the esophageal wall with loss of control at the postganglionic, non-adrenergic and non-cholinergic inhibitory nerves.
  - *Secondary achalasia* is due to parasites, e.g., Chagas disease, malignancy, diabetes, infiltrative, e.g., amyloid, sarcoid, post-vagotomy.
- There is increased risk of *SCC* of the esophagus secondary to chronic stasis.

Presentation and investigation

Progressive dysphagia, regurgitation of undigested food, weight loss, halitosis, chronic aspiration.

- Barium esophagram shows tertiary contractions, bird's beak image at LES.
- Manometry shows normal LES pressure, but failure to relax, no progressive peristalsis, and tertiary contractions.

Medical management

- Oral nitrates and calcium channel blockers.
- Balloon dilation. 1–7% perforation rate. Most patients report symptomatic relief. Half of these will require a second dilatation within 5 years.
- Injection of bothlinum toxin (Botox) into the LES (inhibits Ach release from nerve endings). 85% of patients have symptomatic relief after Botox therapy lasting 3–12 months. 50% recurrence after 6 months.

Surgical options

- ***Myotomy antireflux procedure*** (wrap): main controversies are the extent of the myotomy, need for a wrap, thorascopic versus laparoscopic approach, and management of patients who fail surgery.
- Laparoscopic or robotic ***Heller myotomy*** with 180° wrap:
  - Port set-up is similar to laparoscopic Nissen fundoplication (📖 p779).
  - After mediastinal dissection anterior myotomy is performed through longitudinal and circular layers of esophagus, extending from inferior pulmonary vein to 2–3cm onto anterior stomach.
  - To prevent reflux a 180° wrap is performed.
- Dor anterior wrap Toupet posterior wrap.
- ***Esophagectomy*** (📖 p764) is reserved for end-stage achalasia, sigmoid or megaesophagus, multiple reflux strictures, two or more prior myotomies.

Nutcracker esophagus

- ***Pathology:*** normal peristalsis, but in the distal esophagus contraction amplitudes are up to 180mmHg, and duration >6s.
- ***Presentation:*** extreme atypical chest pain. Emotional triggers.
- ***Investigation:*** manometry shows high-amplitude contractions with contraction durations >6s, normal LES.
- ***Management:*** primarily medical, nitrates and calcium channel blockers.

Diffuse esophageal spasm

- ***Pathology:*** degenerative changes in vagus, muscle hypertrophy. Simultaneous contractions in esophagus leading to 'corkscrew' or 'rosary bead' appearance, and epiphrenic diverticulum.
- ***Presentation:*** substernal atypical pain that is similar to angina, anxiety, high association with psychiatric disorders.
- ***Investigation:*** manometry shows normal or hypertensive LES, normal relaxation of LES; simultaneous, high-amplitude esophageal contractions. Barium esophagram shows simultaneous esophageal contractions.
- ***Treatment:*** nitrates and calcium channel blockers, anxiolytics.
- ***Surgery:*** reserved only for failed medial management: long myotomy via left thoracotomy, myotomy extending from aortic arch onto the LES, with incomplete wrap such as Belsey fundoplication to limit reflux after dividing LES. Good results in only 2/3 of patients.

Scleroderma

- Atrophy of smooth muscle components, fibrous infiltration, and incompetence of LES is found in 80% of patients with scleroderma.
- Manometry shows an incompetent LES and lack of peristalsis.
- Antireflux procedures are not very successful, and may exacerbate dysphagia: occasionally esophagectomy is required.

Hernias and ruptures

Key facts

- Herniation of the body and/or fundus of the stomach through the diaphragmatic esophageal hiatus into the mediastinum, anterior or lateral to the esophagus is considered a paraesophageal hiatal hernia.
- Found in a mostly elderly population.

Pathology

- Hiatus hernia is divided into four types:
  - Type I (sliding hiatal hernia) occurs with migration of the *GEJ above the diaphragmatic hiatus* so reflux is predominant symptom.
  - Type II (rolling or paraesophageal) due to upward migration of the gastric fundus into the posterior mediastinum; however, the *GEJ remains below diaphragm* in normal position within the abdomen.
  - Type III (combination) massive or giant paraesophageal hernia where the majority of the stomach, including the fundus and GEJ have herniated into the mediastinum.
  - Type IV (complex) massive hernia where the stomach and other abdominal organs, such as the spleen, colon, small bowel have herniated into the mediastinum.
- Complications are ***functional*** (GERD, Barrett's, esophagitis, strictures, aspiration) and ***mechanical*** (obstruction, volvulus, and impaired pulmonary function in the case of massive hernias).

Presentation

- Often are asymptomatic.
- Dysphagia, odynophagia, hematemesis, vomiting, anemia, dysphagia, regurgitation, postprandial fullness.
- If a patient complaints of chest or epigastric pain, a NG tube should be placed immediately to decompress the stomach and avoid strangulation.

Investigation

- Barium esophagram defines anatomy, level of obstruction, position of stomach, and emptying of esophagus and stomach.
- EGD helps exclude other pathology, and permits biopsy to assess esophagitis and strictures.
- Esophageal function may be further assessed with manometry and pH.
- Chest CT provides information on contents and size of hernia sac.

Management

- Type I hernias may not require repair: GERD may be managed medically (📖 p778) with antireflux surgery (📖 p787) if medical management fails.
- Type II–IV hernias require surgical repair.

Surgery

- Can be done laparoscopically or through a left thoracotomy.
- The hernia sac and its contents is reduced (via laparoscopic, laparotomy or left thoracotomy approach).
- The sac is excised and the esophageal hiatus narrowed posteriorly with or without mesh reinforcement, to reduce recurrence (permanent mesh is avoided to prevent chronic constriction or perforation).

- If there is any tension when the GEJ is returned into the abdomen, then an esophageal lengthening procedure should be performed, e.g., Collis gastroplasty with Nissen fundoplication (📖 p787).
- A gastropexy may be performed to fix the stomach in the abdomen.
- The mortality of surgery is around 2–5%, higher in emergency repair, and the risk of recurrence around 10–20%.

Esophageal ruptures

General principles are covered on 📖 p790. Specific types include:

Boerhaave syndrome

- Boerhaave syndrome classically presents as sudden onset severe chest pain radiating to back, abdomen left shoulder, worse on swallowing, and starting after repeated and prolonged vomiting.
- In 1723 Hermann Boerhaave detailed a postmortem on Baron Jan Gerrit von Wassenaer, who suffered postemetic esophageal rupture after self-inducing emeses after a heavy meal—he found a transverse tear of the distal esophagus.
- Vomiting is the most common cause, and is thought to be secondary to simultaneous closing of the upper and lower esophageal sphincters, with resultant increase of intraluminal pressure.
- 90% of injuries occur in the lower 1/3 of the esophagus on the left side.
- Unlike originally described, commonly a longitudinal tear (3–10cm).
- Usually in men age 35–55, often occurs in alcoholics.
- Repair of esophageal rupture is described on 📖 p790.

Esophageal foreign body and food impaction

- Foreign body impaction is common in children and psychiatric patients. Children commonly ingest coins. Elderly people may accidentally ingest their dentures. Represents 12% of esophageal perforations.
- Common area of impaction is the broncho-aortic constriction point.
- Meat impaction is dissolved by having the patient swallow 5mL of 20% papain (meat tenderizer) in 10% of alcohol every 5min for 1h.
- Flexible endoscopy or rigid esophagoscopy is often adequate.
- Do not push the impacted object forward, as there may be a distal obstructing lesion, especially if the impacted object is in the distal 1/3 of the esophagus. Pushing against a fixed obstruction may perforate.
- Esophagotomy may be required if the foreign body cannot be removed endoscopically: objects in the distal 1/3 of the esophagus should be approached via a left posterolateral thoracotomy, objects in the proximal 2/3 of the esophagus should be approached via a right posterolateral thoracotomy.

Esophageal mucosal dissection

- Mucosal injury after ingestion of sharp food, e.g., bone or shell, resulting in dissection along the submucosa, causing exquisite odynophagia.
- Thin barium esophagram demonstrates migration of contrast within the wall of the esophagus, but no extravasation.
- Therapy is supportive therapy with pain control. Self-limiting.

Esophageal perforation

Key facts

- Esophageal perforation is a full-thickness injury to the cervical, thoracic, or abdominal esophagus, resulting in extravasation of luminal contents into the neck, mediastinum, pleural space, or abdominal cavity.
- Overall mortality associated with esophageal perforation is 18%.
- Mortality is determined by cause, location, and treatment delay:
  - Spontaneous esophageal perforation carries a 36% mortality; iatrogenic 19% mortality; traumatic 7% mortality.
  - Cervical perforations carry a 6% mortality; thoracic perforation a 27% mortality; abdominal perforations a 21% mortality.
  - Diagnosis <24h, mortality 14%; >24h, mortality 27%.

Pathology

- Instrumental (60–75%), e.g., endoscopy, dilation, balloon and bouginage, intubation errors, laser therapy.
- Non-instrumental: barogenic trauma (15%), postemetic ***Boerhaave syndrome*** (📖 p789) (excess vomiting results in rupture of esophagus).
- Penetrating trauma (9%), operative trauma (2%), caustic injury (📖 p792).
- Swallowed foreign bodies or meat impaction (12%), malignancy (1%).

Presentation

- Symptoms depend on location, common points of perforation occur at three naturally occurring areas of esophageal narrowing (📖 p659):
  - Cervical constriction point: cricopharyngeus muscles and Killian's triangle (14–16cm from the incisors).
  - Mid-esophageal constriction point: bronchoaortic constriction (22–24cm from the incisors).
  - Distal esophageal constriction point: esophagogastric junction (40–45cm from the incisors).
- Cervical perforation: low severity, with contamination limited by neck, but may extend down anterior to the prevertebral fascia ('danger space') into the mediastinum. May present with neck pain, dysphagia, dysphonia, subcutaneous emphysema.
- Thoracic perforation: may rapidly contaminate the mediastinum and the pleural space, resulting in chemical irritation, inflammation, marked fluid shifts, and sepsis. May present with chest pain, tachycardia, fever, leukocytosis, hypotension.
- Abdominal perforation: extravasation of luminal contents into the abdomen. May present with abdominal pain, and a clinical examination demonstrating epigastric tenderness and peritoneal signs.

Investigation

- CXR shows pneumomediastinum and subcutaneous emphysema.
- All stable patients should undergo a thin barium esophagram: water-soluble contrast has poorer sensitivity and poses a risk of pneumonitis if aspirated. If patients are going to the OR for other reasons, and an

esophageal injury is suspected, then a flexible endoscopy should be performed—it may not always identify the lesion.
- Concurrent tracheal injury repairs should be separated from esophageal repairs with viable tissue, such as strap muscle, intercostal muscle, pericardial fat, pleural patch, or omentum.

Operative management

Cervical perforation

- A cervical esophageal perforation is approached via a standard left neck incision, anterior to the sternocleidomastoid muscles, and the carotid–jugular bundle. The omohyoid muscle is divided, and the incision is carried down to the esophagus and anterior spine.
- If a cervical perforation cannot be primarily repaired, leaving the neck open to allow for drainage is adequate, and the injury should heal with packing in 2 weeks.
- The mucosa injury is always more extensive than the muscular injury.
- The muscularis propria (longitudinal and circular) layers are divided with cautery, exposing the entire mucosal defect; devitalized tissue is debrided. The mucosal defect is closed using running 4/0 PDS. The muscular layer is closed with running or interrupted 4/0 PDS and the neck wound is closed in layers over a drain (1/4 inch Penrose).

Intrathoracic perforation

- All intrathoracic injuries in the absence of intrinsic disease, e.g., malignancy should undergo primary repair regardless of time from diagnosis.
- In achalasia (📖 p786), a myotomy should also be performed, whereas in the setting of malignancy, an esophagectomy is necessary.
- Distal 1/3 lesions are reached via a left posterolateral thoracotomy. Middle and upper 1/3 lesions are reached via a right thoracotomy.
- The esophagus is mobilized.
- The mucosal injury is always more extensive than the muscular injury.
- The longitudinal and circular muscle layers are divided with cautery, revealing the entire extent of the mucosal injury. A 48–52F Maloney bougie is passed to prevent narrowing of the esophagus by the repair.
- The submucosal edges are grasped with Allis clamps; the defect is closed with a stapler and reinforced with an intercostal muscle flap.

Esophageal stents

- With small perforations, or patients who are poor surgical candidates, there have been numerous reports documenting success with endoscopically placed temporary covered esophageal stents.
- In the setting of intrathoracic perforation, the pleural space still needs drainage either with tube thoracostomy or by VATS.

Outcomes

- Complications after primary repair of esophageal perforation include leak, empyema, soft tissue infections, recurrent laryngeal nerve injury.
- Overall mortality <10%.

Caustic esophageal injuries

Key facts

- There are 5000 caustic ingestion injuries per year in the USA.
- Occurs in young children and adults attempting suicide.
- Offending agents include acids, alkalis, and household materials such as detergents and bleach.
- Acids and alkali cause reflex pyloric and cricopharyngeal spasm resulting in pooling in the gastric antrum. Contractions in the esophagus and stomach distribute the corrosive agents. As a result can have full tissue destruction of the distal esophagus and the stomach.

Pathology

Strong alkali

- Includes household drain cleaners.
- Alkali causes liquefaction necrosis with deep tissue penetration.

Strong acids

- May relatively spare the esophagus and cause severe gastric injury.
- The depth of tissue injury is often less than that of alkali.

Presentation and investigation

- Thin barium esophagram is helpful in identifying areas of perforation.
- Chest CT.
- Flexible endoscopy is critical in determining extent of mucosal destruction. Mucosal injuries are described similar to burns:
  - *First-degree:* mucosal hyperaemia and oedema.
  - *Second-degree:* mucosal ulceration with blisters and exudates with pseudomembrane formation.
  - *Third-degree:* deep ulcerations with charring and eschar formation, with massive edema and luminal obliteration.

Management

Initial management

- Airway control, fluid resuscitation and broad-spectrum antibiotics.
- No role for prophylactic corticosteroids.

First-degree injury

- Expectant management with observation for 24–48h.

Second-degree injury

- Careful observation for perforation.
- These patients will often need long-term surveillance as stricture formation is common.
- Dilations should not be performed until 6–8 weeks after initial injury.
- Patients with strictures intractable to dilation may require esophagectomy with gastric reconstruction if available, or colon conduit.

Third-degree injury

- Complete necrosis and/or perforation of the esophagus requires esophagectomy with delayed reconstruction.

- A THE is often made easier due to edema in the mediastinum.
- The esophageal hiatus is closed.
- A feeding jejunostomy is placed, and if the stomach is viable, then a draining gastrostomy is performed.
- A diverting end esophagostomy (spit fistula) is performed.
- The residual esophagus is brought out via a standard left neck incision, and tunnelled along the chest and exteriorized as an ostomy just below the clavicle via interrupted 4/0 Vicryl. This allows for a flat surface on which to place an ostomy bag (Fig. 15.5).
- The end esophagostomy should be dilated with a finger every day to prevent stricture formation.

Esophageal discontinuity

- If during attempt at primary repair of the esophagus, the patient is too unstable to move forward, or the tissues are too compromised by sepsis, the area in question can just be stapled closed and adequately drained.
- The patient can be brought back when stable for an end oesophagostomy or 'spit fistula' (Fig. 15.5), and placement of an enteral feeding tube, for long-term nutrition.
- Eventually completion esophagectomy with gastric or colon conduit reconstruction can be performed.

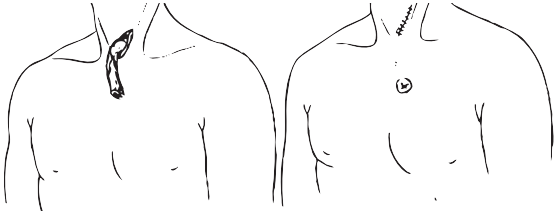


Fig. 15.5 Formation of a 'spit' fistula.

Benign tumors and cysts

Key facts

- Benign esophageal tumors are rare lesions which can appear anywhere within the wall of the esophagus. Most common etiology is leiomyoma, lipomas, and hemangiomas (Table 15.6).
- Esophageal cysts are rare; they may be congenital or acquired.
- Symptoms depend upon tumor location: dysphagia, odynophagia, aspiration, hematemesis, regurgitation of lesion into mouth or airway if pedunculated, atypical chest pain.
- Chest CT and MRI are useful for delineation of anatomy, endoscopy and EUS provides biopsy and histology, and location within wall of esophagus.
- Treatment if required for symptom control is surgical extirpation via myotomy: need for esophagectomy is rare.

Esophageal duplication cysts

Pathology

- These rare congenital cysts result from a developmental failure of the posterior division of the primitive foregut.
- Intramural and enclosed within the esophageal wall, lined with squamous epithelium, though may contain gastric epithelium.
- Frequently found on the right inferior aspect of the esophagus, and do not have continuity with the esophagus.
- May be associated with other congenital abnormalities:
  - Vertebral defects and spinal cord defects.
  - Congenital tracheoesophageal fistula.
  - Esophageal atresia.
  - Other GI visceral duplications.

Presentation

- In children they may obstruct the esophageal lumen or cause respiratory symptoms secondary to compression effect.
- Adults are often asymptomatic. Symptoms include dysphagia, dyspnea, atypical chest pain. Perforation, bleeding, and malignant degeneration is rare.

Diagnosis

- Chest CT and MRI are useful for delineation of anatomy and fluid or viscous contents.
- Biopsy should be avoided.
- Endoscopy and EUS rules out intraluminal pathology, and confirms intramural location.

Treatment

- Surgical extirpation via a myotomy.

Other congenital cysts

Bronchogenic esophageal cysts

These congenital intramural cysts are found in the distal 1/3 of the esophagus, and contain cartilage.

Esophageal gastric cysts
- These intramural cysts are lined with gastric mucosa.
- May present with a peptic ulcer.

Inclusion cysts
Intramural. Contain respiratory or squamous epithelium.

Acquired cysts
- Retention cysts.
- May be single or multiple.
- Found in the upper 1/3 of the esophagus.

Leiomyoma
- Most common benign tumor of the esophagus.
- Usually found in the middle and lower 1/3 of the esophagus.
- Intramural location.
- 2:1 male to female ratio, rare in children.
- Usually single intraluminal polyp found in the cervical esophagus.

Lipoma
- Commonly presents as an intraluminal polyp, in the upper esophagus.
- Can grow to large size and may present as a mass in the throat which disappears upon swallowing.

Haemangioma
- Submucosal lesion.
- Usually asymptomatic. Symptoms include atypical chest pain, dysphagia, and hematemesis.

Table 15.6 Location of benign tumors and cysts within the esophageal wall

Lesion	Location
Squamous papilloma	Mucosa
Fibrovascular polyp	
Retention cyst	
Lipoma	Submucosa
Fibroma	
Neurofibroma	
Granular cell tumor	
Hemangioma	
Salivary gland type tumor	
Leiomyoma	Muscularis
Duplication cyst	
Adventitia	
Foregut cyst	

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